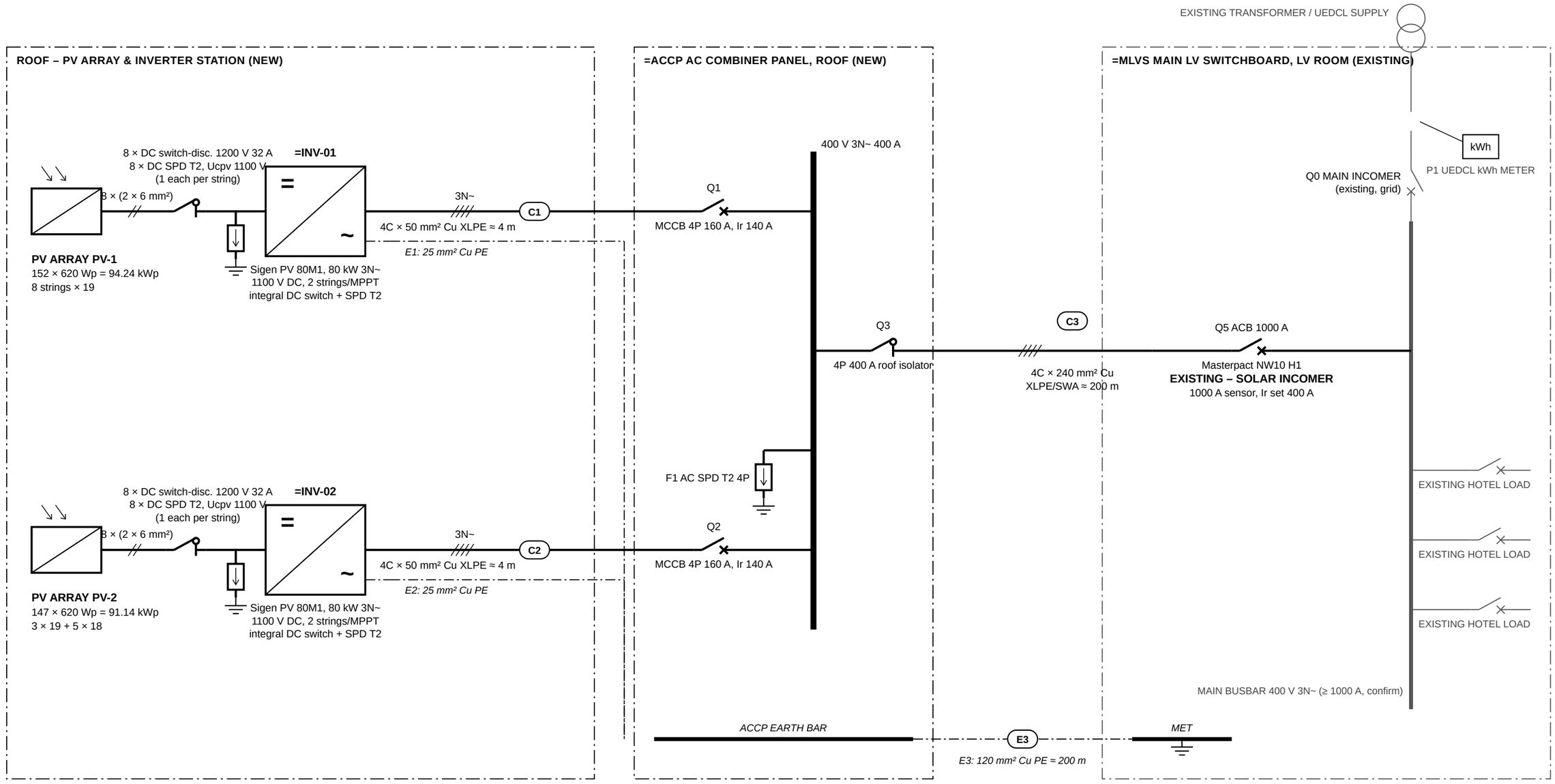


SYMBOLS (IEC 60617)

- PV array / PV generator
- Inverter DC/AC
- Circuit breaker (MCCB / ACB)
- Switch-disconnector (isolator)
- Surge protective device (SPD)
- Earth / main earthing terminal
- Heavy = new, light = existing; strokes = conductors
- DC conductor · PE conductor

CABLE SCHEDULE

TAG	FROM → TO	CABLE	LENGTH	PROTECTION
S1–S16	PV strings → inverters	2 × 6 mm² Cu H1Z2Z2-K	≈ 40 m	DC isolator 32 A + DC SPD T2
C1, C2	INV-01/02 → ACCP Q1/Q2	4C × 50 mm² Cu XLPE	≈ 4 m	MCCB 160 A (Ir 140 A)
C3	ACCP Q3 → MLVS Q5	4C × 240 mm² Cu XLPE/SWA	≈ 200 m	ACB NW10 H1, Ir 400 A
E1, E2	INV-01/02 → ACCP earth bar	25 mm² Cu (G/Y)	≈ 4 m	–
E3	ACCP earth bar → MET	120 mm² Cu (G/Y)	≈ 200 m	–
–	Array bonding / spine	16 mm² / 35 mm² Cu (G/Y)	≈ 280 m	–

Lengths indicative – confirm on site. Sizes per IEC 60364-5-52 / 5-54 (see design report).

NOTES

1. 299 × 620 Wp = 185.4 kWp DC; 2 × 80 kW = 160 kW AC; DC/AC 1.16.
2. Grid-tied, no battery. DC SPD T2 + DC isolator on every string.
3. C3 200 m: VD ≈ 2.0 % at 232 A. Q5 = existing NW10 H1, Ir 400 A.
4. Final ratings to equipment datasheets and detailed design.

CLIENT: **Kampala Marriott Hotel**	PROJECT: **Kampala Marriott – Rooftop PV 185.4 kWp**

TITLE: **SINGLE-LINE DIAGRAM – PV SYSTEM & GRID CONNECTION**

DRAWING No.	REV	SCALE / SIZE	DATE
PV-SLD-D18102-004	**B**	**NTS / A3**	**09.11.2026**

DRAWN	CHECKED	APPROVED